

SYNAPSE

A Computational Disease Dynamics Company

DOCUMENT	3-Year Strategic Roadmap & North Star
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"We are building the infrastructure to model disease as a learnable, intervention-aware dynamical system — starting with diabetic retinopathy progression, expanding toward a general computational disease dynamics platform."

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Part I — Foundation

1. Executive Summary & Honest Probability Assessment

Synapse is a computational disease dynamics company. We are building the first intervention-aware, multimodal, latent state-space modeling system for disease progression — starting with diabetic retinopathy and expanding toward a general platform for understanding how diseases evolve through time under treatment.

This document is the north star. Every tactical decision, sprint plan, and execution document references back to this. It is the constitution.

The One-Sentence Mission

We model disease as a hidden dynamical system observed through noisy signals and modified by intervention, with the goal of forecasting trajectories, simulating treatment effects, and eventually generating clinically valid synthetic populations.

Honest Probability Assessment

We will not pretend this is easy. For a solo founder with limited prior ML and ophthalmology experience, building toward a deep-tech disease dynamics platform, the realistic probability distribution looks like this:

Outcome	Probability	Timeline
Build a working prototype in 6 months	70%	Month 6
Produce a publishable scientific result	40%	Month 9-12
Validate the core thesis on real data	20%	Month 12-18
Become a \$100M+ company	5-10%	Year 4-5
Achieve unicorn (\$1B+) status	2-5%	Year 5-7

Critical reframe: A 2-5% probability of a billion-dollar outcome is still an extraordinary bet. We are not pretending the path is easy or certain. We are betting that the category — computational disease dynamics — is large enough that even a small probability of success justifies the work.

Our mission for the first 6 months is not to become a unicorn. It is to **earn the right to pursue the unicorn outcome** by producing credible scientific evidence.

What Success Looks Like (Honest Version)

Month 6 — We have built a reproducible pipeline using RETFound + latent state-space modeling. We have benchmarked against strong baselines on EyePACS image classification. We have a preprint-quality writeup. The result is suggestive but not definitive.

Month 12 — We have JEPA-style fine-tuning working. We have access to at least one longitudinal dataset. We have early conversations with academic partners. We have validated something — even if not the full thesis.

Year 2 — We have controlled longitudinal data. We have demonstrated trajectory prediction on real progression. We have first paid pilot or grant. We have raised seed capital or have credible offers.

Year 3 — We have multimodal HAWVA working. We have a second disease in pilot. We have first pharma conversation. We have Series A or strong revenue traction.

2. Vision, Mission, and Long-Term Ambition

The 10-Year Vision

Synapse becomes the infrastructure layer for computational disease understanding. Pharmaceutical companies use Synapse to design better trials. Biotech companies use Synapse to develop rare disease treatments. Researchers use Synapse to model disease evolution. Clinicians eventually use Synapse-derived tools for prognosis and treatment planning. The same underlying engine — disease as a learnable, intervention-aware dynamical system — powers all of it.

If AlphaFold gave us a computational understanding of protein structure, Synapse aims to give us a computational understanding of disease evolution. That is the ambition. We do not claim we will reach it. We claim it is worth attempting.

The Core Insight

Medicine stores observations: scans, notes, labs, diagnoses, treatments. But medicine does not explicitly model the underlying disease process itself. Doctors implicitly perform Bayesian inference — they observe noisy signals, estimate hidden disease state, apply interventions, and update beliefs. We are formalizing that process computationally.

Observations are not the disease. The disease is hidden. Our job is to infer the hidden state from observations and model how it evolves.

Values & Principles

Scientific rigor over marketing claims. We do not overclaim. Every assertion must be backed by evidence. Every uncertainty must be quantified.

Build the foundation before the cathedral. Discipline in execution order. The ambition stays vast. The execution stays focused.

Honest probability over false confidence. We tell investors, partners, and ourselves the truth about what we can and cannot prove.

Public science where possible. We publish. We open-source non-core code. Credibility is the real moat.

Mission alignment with patient outcomes. We are not here to make money first. We are here to help understand disease. Money follows.

3. The Scientific Hypothesis (Falsifiable Core)

Every serious deep-tech company has a falsifiable scientific hypothesis at its core. If the hypothesis is true, the company is real. If it is false, the company should pivot or fold. Our hypothesis is:

H1: *Disease progression can be modeled as a learnable latent dynamical system, where hidden disease state evolves over time according to learnable transition functions, and observations (imaging, labs, clinical signals) are noisy projections of that hidden state.*

H2 (extension): *When this latent state model is augmented with treatment history and physician policy modeling, it can recover intervention-aware dynamics that enable counterfactual reasoning (what would have happened with different treatment).*

H3 (extension): *A validated disease dynamics model can generate synthetic longitudinal cohorts that preserve causal structure, enabling research, trial design, and rare disease investigation.*

How We Test H1 (Year 1)

Test: Can a latent state-space model (RETFound + latent ODE) outperform strong baselines (GRU-D, T-LSTM, tabular survival) on diabetic retinopathy progression forecasting?

Success criteria:

- Model beats baselines by $\geq 10\%$ on at least one clinically meaningful endpoint
- Performance generalizes externally (Messidor-2, temporal split) with $< 15\%$ degradation
- Predicted uncertainty is calibrated against observed outcomes
- Latent representations cluster meaningfully by disease severity (validated qualitatively)

Failure criteria:

- Baselines match or exceed our model
- Performance collapses on external validation ($> 30\%$ degradation)
- Uncertainty estimates are uncalibrated
- Latent representations encode dataset artifacts (camera, site) rather than disease

Critical Caveats

We must be honest about what Year 1 can and cannot prove. EyePACS is predominantly cross-sectional. Without controlled longitudinal data, our Year 1 claim is bounded by what the data supports. We will likely demonstrate:

Claim	Provability
✓ Strong image-based DR classification with pretrained encoders	Provable in Year 1
✓ Competitive performance on image-to-severity tasks	Provable in Year 1

? Suggestive evidence of latent dynamics structure	Partially provable with caveats
✗ Definitive proof of disease dynamics learnability	Requires Year 2 + longitudinal data
✗ Causal treatment effect estimation	Requires Year 2 + controlled data
✗ Validated synthetic cohort generation	Requires Year 3 + validated dynamics

4. What Makes This Different

The retinal AI market is crowded for screening. IDx-DR, EyeArt, AEYE Health, and many others perform diabetic retinopathy detection. Foundation models like RETFound exist. Synthetic medical data vendors exist. We are not competing in those categories.

Our lane is different: we ask where disease is going, not what it is right now.

Category	Status	Our Position
DR image screening	Crowded, FDA-cleared	We don't compete — we model future progression
Retinal foundation models	Mature (RETFound, etc.)	We use them, not rebuild them
Synthetic EHR generation	Commodity	Synthetic data emerges from our model, isn't the product
Disease simulation (general)	Niche, mostly oncology	We expand it to imaging-rich progression diseases
Intervention-aware dynamics	Open territory	This is our lane

The Five Differentiators

1. We model time, not snapshots. Most retinal AI answers 'what is this eye now?' We answer 'where is this eye going?' That temporal framing is the wedge.

2. We model intervention, not just observation. Eventually our system will reason about counterfactuals: what would happen with different treatment? This is what pharma actually needs.

3. We treat observations as noisy, not as truth. EHRs, scans, labs are projections of hidden state. We infer the state, not the labels. This is a more honest scientific framework.

4. We pursue scientific credibility before product. We publish before we sell. Our first asset is peer-reviewed evidence, not a SaaS dashboard.

5. We aim at the right industry. Pharma R&D; (\$200B/year) and biotech rare disease (\$50B+/year) are our customers — not direct-to-consumer or clinical workflows.

Part II — Architecture

5. The Full Synapse Architecture (All Tiers)

The complete Synapse system has seven tiers. Not all are built immediately. The MVP is Tiers 1-4 (minimal). Tiers 5-7 emerge in Years 2-3. This section documents the full architecture for reference — execution documents will describe what is actually being built at each phase.

Tier 1 — Data Ingestion & Normalization

Purpose: Ingest asynchronous, sparse, irregular medical observations and normalize them into standard formats.

Components:

- Image loaders (DICOM, JPEG, PNG)
- EHR parsers (FHIR, HL7, custom)
- Temporal aligners (handle irregular timestamps)
- Missing data handlers (imputation, masking)
- Quality validators

Built in MVP (basic version)

Tier 2 — Encoder Layer (Representation Learning)

Purpose: Convert raw observations into latent embeddings using pretrained backbones plus task-specific fine-tuning.

Components:

- RETFound retinal encoder (pretrained, frozen initially)
- JEPA-style fine-tuning (Year 1+, on top of RETFound)
- Clinical encoder (labs, demographics, treatments)
- Temporal encoder (visit spacing, time features)
- Modality-specific encoders for future expansion

RETFound only in MVP; JEPA fine-tuning Y1+

Tier 3 — HAWVA Belief State (Multimodal Fusion)

Purpose: Maintain a persistent probabilistic belief over hidden disease state, updated as new observations arrive.

Components:

- Uncertainty calibration (MVP)
- Simple multimodal fusion (Year 1)
- Bayesian evidence aggregation (Year 2)
- Cross-attention fusion (Year 2+)
- Full Bayesian filter (Year 3)

Minimal (uncertainty only) in MVP

Tier 4 — Disease Dynamics Engine

Purpose: Model how disease state evolves through time under intervention. The scientific core.

Components:

- Latent ODE model (MVP)
- Neural state-space model (alternative)
- Markov / CT-HMM baseline (interpretability)
- Treatment-conditioned transitions (Year 2)
- Physician policy separation (Year 2)
- Stochastic differential equations (Year 2+)

Latent ODE in MVP; intervention awareness Y2+

Tier 5 — Active Inference & Decision Support

Purpose: Recommend optimal next observations to reduce uncertainty. Close the loop between model and clinic.

Components:

- Information gain estimation
- Test recommendation engine
- Cost-aware sampling
- Clinical decision support API
- Triage and prioritization

Not in MVP. Year 2-3 capability.

Tier 6 — Simulation & Synthetic Generation

Purpose: Run virtual trials, generate synthetic patient populations, simulate counterfactual interventions.

Components:

- Trajectory forecasting engine

- Counterfactual simulator
- Synthetic cohort generator
- Virtual trial designer
- Population modeling tools

Not in MVP. Year 2-3 capability.

Tier 7 — Products & APIs

Purpose: Customer-facing interfaces — APIs, dashboards, integration tools — for pharma, biotech, and research customers.

Components:

- REST/GraphQL API
- Python SDK
- Dashboard UI
- Pharma integration tools
- Documentation portal
- Audit logging

Not in MVP. Year 2-3 capability.

6. Component Inventory: What Exists, What's Future

This is the complete inventory of every component, module, and capability that Synapse will eventually contain. It functions as both a roadmap reference and a checklist. Components are organized by tier and labeled with their status.

Component	Tier	Status	Target
Image data loader	1	MVP	Month 1-2
EHR/clinical data parser	1	MVP	Month 2-3
Temporal alignment system	1	MVP	Month 2-3
RETFound integration	2	MVP	Month 1-2
JEPA fine-tuning layer	2	Year 1+	Month 7-9
Clinical/demographic encoder	2	Year 1+	Month 8-10
Latent ODE model	4	MVP	Month 3-5
Baseline models (GRU-D, T-LSTM, Cox)	4	MVP	Month 3-4
Uncertainty calibration	3	MVP	Month 5-6
Multimodal fusion (simple)	3	Year 1+	Month 9-12
Treatment-conditioned model	4	Year 2	Month 13-18
Physician policy model	4	Year 2	Month 16-20
Bayesian belief fusion	3	Year 2+	Month 18-24
Active inference engine	5	Year 2-3	Month 20-30
Trajectory forecasting API	6	Year 2+	Month 18-24
Synthetic cohort generator	6	Year 3	Month 24-30
Virtual trial simulator	6	Year 3	Month 27-33
REST API	7	Year 2+	Month 15-20
Python SDK	7	Year 2+	Month 18-24
Customer dashboard	7	Year 3	Month 24-30
AMD disease model	4	Year 3	Month 24-30
Glaucoma disease model	4	Year 3+	Month 30-36

7. Technical Stack & Tooling

Our stack is selected for two priorities: scientific credibility (must be the right tools for the research) and execution speed (must work well with AI-assisted development). We optimize for boring, well-supported choices where possible.

Languages & Frameworks

Layer	Technology	Rationale
Primary language	Python 3.11+	Standard for ML; massive ecosystem
ML framework	PyTorch	Industry standard; better for research
ODE solver	torchdiffeq	Battle-tested for Neural ODEs
Probabilistic ML	Pyro / NumPyro	Bayesian modeling, future HAWVA
Image processing	Pillow, OpenCV	Standard libraries
Data handling	pandas, polars, numpy	Standard data tools
Experiment tracking	Weights & Biases	Free tier sufficient for solo founder
Version control	Git + GitHub	Public repository for credibility
Cloud compute	Google Colab → GCP/AWS	Free tier first, scale later
Development env	VS Code + Claude Code	AI-accelerated development
API framework	FastAPI	Year 2+ when productizing
Database	PostgreSQL	Year 2+ for production data

Data Sources

Dataset	Access	Purpose
EyePACS (Kaggle)	Public	~35K retinal images, severity-labeled. Primary MVP dataset.
Messidor-2	Public	~1.7K images, external validation.
UK Biobank	Controlled access	100K+ participants with imaging + outcomes. Year 1+ target.
Moorfields DR Dataset 005	Licensed	45K patients, 10+ years follow-up. Year 1-2 target.
Moorfields DMO (Dryad)	Public	2K patients with anti-VEGF treatment outcomes. Year 1-2.
DRCR Retina Network	Application	Protocolized trial data. Year 2 target.
RETFound pretrained weights	Public	Available via Hugging Face. Day 1.

Part III — Execution

8. Year 1 — Foundation & Scientific Credibility

Year 1 is about earning the right to pursue this vision. We are not building a company yet. We are building scientific evidence that the company can exist. The deliverable is credibility — a working prototype, a preprint, reproducible code, and clarity on whether the thesis is real.

Year 1 North Star Metric

By Month 12, we have produced peer-review-quality evidence that latent state-space models can forecast diabetic retinopathy progression competitively with strong baselines, and we have at least one academic partnership or seed funding offer in motion.

Q1 (Months 1-3): Foundation

Theme: Data, environment, baselines

Objectives:

- Complete data pipeline for EyePACS + Messidor-2
- Integrate RETFound encoder; produce embedding dataset
- Build all baseline models (logistic regression, GRU-D, T-LSTM, Cox survival)
- Set up reproducible experimentation infrastructure (W&B, Git, README)
- Initial GitHub repo with clean structure

Deliverable: Reproducible baselines on EyePACS + project infrastructure

Q2 (Months 4-6): Core Model

Theme: Latent state-space model + first results

Objectives:

- Implement Latent ODE model on retinal embeddings
- Train and tune on EyePACS longitudinal subset
- Add uncertainty quantification (MC dropout or variational)
- External validation on Messidor-2 (with appropriate caveats)
- Temporal split validation on EyePACS
- Draft preprint outline + early figures

Deliverable: Working state-space model with results vs baselines

Q3 (Months 7-9): JEPA & Expansion

Theme: Add predictive fine-tuning + multimodal beginnings

Objectives:

- Implement JEPA-style fine-tuning on top of RETFound
- Compare JEPA-fine-tuned vs frozen RETFound on progression tasks
- Begin clinical encoder (synthetic risk factors if no real data)
- Reach out to potential academic partners (Moorfields, DRCR, UCSF)
- Apply for controlled-access datasets
- Write preprint draft

Deliverable: Preprint draft + partnership conversations

Q4 (Months 10-12): Validation & Outreach

Theme: Publish, validate, position

Objectives:

- Submit preprint to arXiv
- Submit conference/journal paper (NeurIPS, MICCAI, JAMA Ophthalmology)
- If controlled data secured: begin longitudinal validation
- Create pitch deck for investors / partners
- First investor conversations (10+ meetings)
- Decide: continue solo, recruit co-founder, raise seed, or pivot

Deliverable: Preprint published, decision on next phase made

Year 1 Go / No-Go Decision (Month 12)

GREEN LIGHT — Proceed to Year 2 with funding/team building:

- ✓ Model demonstrates measurable advantage over baselines
- ✓ External validation holds within reasonable degradation
- ✓ Preprint receives positive feedback / publication acceptance
- ✓ At least one serious partnership conversation in motion
- ✓ Clear plan for controlled longitudinal data access

YELLOW LIGHT — Iterate before scaling:

- Results are mixed but suggestive
- Some baselines beat the model, others don't
- External validation degrades more than expected
- Limited partnership traction

RED LIGHT — Pivot or fold:

- ✗ Model is consistently inferior to baselines
- ✗ Performance collapses externally
- ✗ No partnership interest after sustained outreach
- ✗ Thesis appears falsified — disease dynamics not learnable from this data

9. Year 2 — Validation & Multimodal Expansion

Year 2 assumes Year 1 produced green-light results. We move from prototype to validated system. We acquire real longitudinal data. We add multimodal fusion. We make the first business steps — partnerships, possibly seed funding, possibly first co-founder or hire.

Year 2 North Star Metric

By Month 24, we have validated latent disease dynamics on real longitudinal clinical data with a partner, demonstrated multimodal fusion improves forecasting, and have either seed funding closed or significant pharma/biotech pilot in motion.

Q5 (Months 13-15): Real Data, Real Validation

- Integrate controlled longitudinal dataset (Moorfields, DRCR, or equivalent)
- Replicate Year 1 results on real longitudinal data
- Build proper multimodal HAWVA (image + clinical + temporal fusion)
- Begin physician policy modeling (where treatment data permits)
- Recruit first technical co-founder or senior advisor

Q6 (Months 16-18): Intervention Awareness

- Implement treatment-conditioned transition model
- Validate intervention effects against known clinical literature
- Causal inference techniques (G-computation, propensity scoring)
- Build first API endpoints for partners
- Close seed round if path is clear (\$500K–\$2M)

Q7 (Months 19-21): Productization & Partnership

- First paid pilot or grant-funded partnership (pharma or biotech)
- Build basic dashboard for partner to use
- Begin AMD disease model (second disease validation)
- Hire 1-2 people if funded (ML engineer + clinical data engineer)
- Submit second-stage paper (treatment-aware modeling)

Q8 (Months 22-24): Scale Preparation

- AMD model in pilot
- Robust API + SDK for partners
- Active inference layer (recommend next-best test)

- Begin Synthetic cohort generation R&D; (not productized)
- Series A preparation or expanded partnership negotiations

Year 2 Critical Risks

- Data access falls through — controlled longitudinal data is hard to secure
- Treatment effect estimation fails to validate against trial data
- Multimodal fusion adds complexity without improving performance
- Pharma partnerships move too slowly to support runway
- Solo founder burnout (or insufficient progress without team)

10. Year 3 — Platform & Pharma Partnerships

Year 3 transitions Synapse from research prototype to platform. We have multiple diseases. We have validated dynamics. We have real customers paying for forecasting and simulation. We are preparing the infrastructure for true scale.

Year 3 North Star Metric

By Month 36, we have a multi-disease platform with at least 2-3 validated disease models, \$1-5M ARR or comparable grant funding, 5-10 team members, and clear Series A momentum or strategic partnership trajectory.

Platform Maturity

- Multi-disease engine (DR + AMD validated; Glaucoma in pilot)
- Full HAWVA Bayesian fusion across modalities
- Active inference for clinical decision support
- Validated synthetic cohort generation
- Counterfactual simulation engine

Business Development

- 2-3 pharma partnerships in active engagement
- First commercial revenue (\$1-5M ARR target)
- Series A fundraising (\$10-25M) or strategic partnership
- Clinical advisory board established
- Regulatory pathway analysis complete

Scientific Output

- 3-5 peer-reviewed publications
- Open-source benchmarks released
- Recognized as category leader in disease dynamics
- Conference talks at top venues (NeurIPS, MICCAI, ARVO)
- Academic partnerships with 2+ major centers

Team & Infrastructure

- 5-10 person team (ML, clinical, engineering, BD)
- Production-grade infrastructure
- Data partnerships formalized

- SOC 2 compliance or equivalent security posture
- Clear regulatory roadmap for any clinical applications

11. AI-Accelerated Execution Model

Building with AI fundamentally changes the solo-founder calculus. Traditional deep-tech timelines assume teams of 3-10 people. With AI-assisted development (Claude, GPT, Copilot, Cursor, etc.), a disciplined solo founder can compress certain phases significantly — but not all phases.

What AI Accelerates

Code generation. Boilerplate, data pipelines, model architecture, training loops — 5-10x speedup possible

Debugging. AI catches bugs and suggests fixes faster than solo debugging

Documentation. Generating docstrings, READMEs, technical docs is near-instant

Literature review. AI summarizes papers, identifies relevant work, suggests methods

Experimentation setup. Hyperparameter sweeps, benchmark scripts, evaluation code

Writing. Preprints, blog posts, investor pitches — 3-5x speedup with good prompting

What AI Does NOT Accelerate

Data acquisition. Getting controlled datasets requires human relationships and legal agreements

Clinical validation. Reviewing model outputs with real clinicians takes their time, not yours

Scientific judgment. Choosing what to validate, what claims to make — still requires human thinking

Trust building. Pharma partnerships, investor relationships, advisor recruitment — human work

Real-world testing. Models work or don't work on real data; AI can't accelerate physics

Regulatory navigation. FDA, IRB, HIPAA — these processes have fixed external pace

Patience and resilience. Six months of work followed by ambiguous results is still six months

The AI-Accelerated Solo Founder Discipline

Three principles guide effective use of AI in this project:

1. Use AI for breadth, human judgment for depth. AI generates 10 options; you pick the right one. AI writes code; you understand it.

2. Verify everything. AI hallucinates. Every claim, every link, every reference must be independently verified. Especially in science.

3. Maintain scientific rigor. AI will let you build a beautiful model that doesn't generalize. Discipline in evaluation and validation must come from you, not the AI.

Part IV — Business

12. Customer Segments & Revenue Model

Synapse sells to organizations that develop drugs, design clinical trials, or conduct disease research. We are B2B infrastructure, not B2C product. Our customers have substantial budgets and clear ROI for tools that accelerate drug development.

Large Pharma

Examples: Roche, Novartis, J&J, Bayer, Lilly, Pfizer

Use case: Trial design, real-world evidence, drug development acceleration

Pricing model: \$500K – \$5M annual licenses; per-simulation fees

Timeline: Year 2-3 first conversations, Year 3-4 first deals

Biotech (Rare Disease)

Examples: Small/mid-cap biotech with rare disease pipelines

Use case: Synthetic cohort generation, virtual trials, regulatory submissions

Pricing model: \$100K – \$1M per project; revenue share on outcomes

Timeline: Year 3+

Contract Research Organizations (CROs)

Examples: IQVIA, ICON, PRA Health Sciences

Use case: Bundle Synapse capabilities into trial design services

Pricing model: Wholesale licensing; revenue share

Timeline: Year 3-4

Academic Medical Centers

Examples: Moorfields, UCSF, Mass Eye and Ear, Johns Hopkins

Use case: Disease research, publication, grant-funded studies

Pricing model: Heavily discounted or free for credibility

Timeline: Year 1-2 (relationship building first)

Regulators & Health Authorities

Examples: FDA, EMA, CMS

Use case: Real-world evidence, post-market surveillance

Pricing model: Long-term, complex

Timeline: Year 4-5+

Revenue Streams (Mature State, Year 5+)

Stream	Description	% of revenue (est)
Platform licensing	Annual pharma/biotech contracts	50-60%
Per-use services	Simulations, cohort generation	15-25%
Custom modeling	Bespoke disease modeling projects	10-15%
Grants & partnerships	Academic and research funding	5-10%
Outcome-based fees	Revenue share on successful drugs	5-10%

13. Funding Path & Capital Strategy

Synapse follows a deep-tech funding trajectory: bootstrap with public data and AI assistance to produce evidence, then raise capital tied to scientific milestones. We optimize for non-dilutive funding (grants) early and dilutive capital only when we have evidence to justify valuation.

Bootstrap (Months 1-6)

Amount: \$0

Source: Solo founder time + AI tools + free compute

Required milestones: Working prototype, preprint, GitHub repo

Implied valuation: N/A

Pre-seed / Grants (Months 6-12)

Amount: \$50K – \$250K

Source: NIH SBIR, NEI grants, ARPA-H, foundations

Required milestones: Validated baseline + initial state-space model

Implied valuation: N/A (non-dilutive preferred)

Seed (Months 12-18)

Amount: \$500K – \$2M

Source: Deep-tech VCs (Lux, DCVC Bio), bio angels

Required milestones: Published paper + partnership conversations + path to data

Implied valuation: \$5M – \$15M

Series A (Year 2-3)

Amount: \$5M – \$15M

Source: Larger deep-tech / bio VCs, strategic pharma

Required milestones: Real-data validation + first paid pilot + multimodal working

Implied valuation: \$20M – \$80M

Series B (Year 3-4)

Amount: \$20M – \$50M

Source: Growth VCs, strategic pharma investment

Required milestones: Multi-disease platform + meaningful revenue

Implied valuation: \$100M – \$500M

Series C / Exit (Year 5-7)

Amount: \$50M+ or acquisition

Source: Late-stage VCs, IPO, or pharma acquisition

Required milestones: Category leader + significant revenue + clear unit economics

Implied valuation: \$500M – \$5B

Non-Dilutive Funding Priorities

Before raising venture capital, we should aggressively pursue non-dilutive funding. This preserves ownership and demonstrates external validation:

- NIH SBIR Phase I (\$300K) — National Eye Institute interest area
- NIH SBIR Phase II (\$1.5M) — follow-on to Phase I
- ARPA-H research grants — health innovation
- American Diabetes Association research funding
- Breakthrough T1D (formerly JDRF) — diabetes complications
- Research Foundation grants — Moorfields, RPB, etc.
- AI-for-health competitions and prizes

14. Go-To-Market & Partnerships

Our GTM is research-led, not sales-led. We earn customer trust through published science, then convert that into pilots and partnerships. We do not have a sales team. We have scientific credibility, partnerships, and word-of-mouth among researchers and clinicians.

Phase 1: Credibility (Year 1)

- Publish preprint and conference papers
- Open-source benchmarks and reproducible code
- Build presence at ARVO, NeurIPS, MICCAI, IEEE
- Engage with retinal AI research community on Twitter/LinkedIn
- Cold outreach to academic labs for collaboration

Phase 2: Pilots (Year 2)

- Academic collaborations with Moorfields, UCSF, Mass Eye and Ear, or equivalent
- Grant-funded research partnerships
- Free or low-cost pilots with retinal specialists
- Begin pharma conversations through advisors / connections
- Build clinical advisory board (3-5 retinal specialists)

Phase 3: Commercial (Year 3)

- First paid pilots with pharma (DR or AMD trial design support)
- Multi-year contracts with 2-3 pharma companies
- Expand into adjacent diseases (AMD, glaucoma)
- Hire BD lead with pharma relationships
- Begin CRO partnerships for trial enrichment services

Key Partnerships (Priority Order)

Moorfields Eye Hospital / INSIGHT. World-class retinal data + RETFound origin; ideal validation partner

DRCR Retina Network. Gold-standard DR trial data; partnership = automatic credibility

UCSF Ophthalmology. Strong AI-for-eyecare program; academic validation

Roche / Genentech. Major DR/DME therapeutic player; obvious commercial fit

Novo Nordisk / Bayer. Diabetes-focused pharma; broader DR interest

ARPA-H OCULAB. Strategic alignment with eye-as-biomarker thesis

Part V — Risk & Reality

15. Risks, Failure Modes, & Mitigation

Every deep-tech company faces a portfolio of risks. The job of the founder is not to eliminate them — that's impossible — but to understand them clearly, monitor them honestly, and have mitigation plans. Below are the risks we explicitly track.

Scientific Risks

Risk	Severity	Mitigation
Latent state-space models don't generalize	HIGH — fundamental to the idea	Validate early and externally; pivot if Month 6 results are weak
Public data lacks sufficient longitudinal structure	MEDIUM-HIGH	Pursue controlled data access (Moorfields, DRCR) by Month 6-9
Models learn dataset artifacts, not disease	HIGH	External validation; subgroup analysis; clinician adjudication
Treatment effect estimation fails causal validation	HIGH (Year 2+)	Use protocolized trial data; honest uncertainty about causal claims
Disease dynamics aren't actually learnable from observations	HIGH (theoretically)	Falsify or pivot — accept this is possible

Technical Risks

Risk	Severity	Mitigation
Solo founder can't execute ML complexity	MEDIUM	AI-assisted development; recruit technical co-founder by Year 2
Compute costs scale faster than runway	LOW (initially)	Stay on free tier as long as possible; cloud credits via accelerators
Model architecture choices age poorly	LOW	Stay method-agnostic; don't brand around specific techniques
Reproducibility issues hurt credibility	MEDIUM	Rigorous version control, environment specs, documented experiments

Business Risks

Risk	Severity	Mitigation
Pharma sales cycles too long for runway	HIGH	Non-dilutive funding bridge; grant pursuit early
Crowded category we didn't see coming	MEDIUM	Constant market monitoring; published papers as defensible moat
Regulatory shifts affecting medical AI	MEDIUM	Track FDA SaMD guidance; position as research tool initially
Investor disinterest in healthcare deep-tech	MEDIUM	Focus on deep-tech and bio-specific VCs
Inability to recruit clinical advisors	MEDIUM	Build credibility through publications first

Personal Risks

Risk	Severity	Mitigation
Founder burnout from solo execution	HIGH	Sustainable pace; community engagement; honest progress tracking
Financial pressure forcing premature pivot	HIGH (no runway)	Day job preservation; grant funding; rapid iteration to evidence
Loss of motivation if results are ambiguous	MEDIUM	Pre-committed go/no-go criteria; honest assessment
Isolation reducing scientific quality	MEDIUM	Active community engagement; advisor cultivation

16. Decision Gates & Pivot Triggers

At specific moments, we evaluate the project honestly and decide whether to continue, iterate, or pivot. These gates are pre-committed — we do not move them to avoid hard decisions. Each gate has clear criteria.

Gate 1 — Month 3

Question: Can we replicate baselines and run experiments reproducibly?

Continue if: All baselines working; reproducible pipeline; clean GitHub repo

Iterate if: Some baselines unstable; pipeline mostly working

Pivot if: Cannot get basic infrastructure working; technical debt overwhelming

Gate 2 — Month 6

Question: Does our latent state-space model show advantages over baselines?

Continue if: Model beats baselines by $\geq 10\%$; external validation reasonable; preprint draft started

Iterate if: Mixed results; some metrics good, others not; investigate why

Pivot if: Baselines consistently match or beat model; investigate alternative approaches

Gate 3 — Month 12

Question: Have we earned the right to pursue this seriously?

Continue if: Preprint published; partnership conversations active; data access path clear; funding interest

Iterate if: Results published but feedback mixed; need more validation

Pivot if: No partnership traction; no funding interest; results don't justify continued investment

Gate 4 — Month 18

Question: Does real longitudinal data confirm our thesis?

Continue if: Validated on controlled data; treatment-aware modeling working; first revenue or grants

Iterate if: Some validation success; need more data or different approach

Pivot if: Thesis doesn't hold up on real data; consider narrower product or major pivot

Gate 5 — Month 24

Question: Are we building a real business or a research project?

Continue if: First paid pilot; multimodal working; team forming; Series A discussions

Iterate if: Research solid but commercial traction slow; explore alternative customers

Pivot if: Cannot achieve commercial momentum; consider acquisition, academic spinout, or fold

Gate 6 — Month 36

Question: Is the platform thesis working?

Continue if: Multi-disease platform; pharma partnerships; clear path to scale

Iterate if: One disease working but expansion harder than expected

Pivot if: Platform thesis weak; consider focused single-disease company or strategic exit

17. Success Metrics & OKRs

We measure what matters. Metrics fall into three categories: scientific (does the model work?), business (is anyone paying?), and execution (are we shipping?). Each quarter has explicit OKRs derived from this framework.

Scientific Metrics

Model performance vs baselines. Concordance index, AUC, accuracy on held-out sets

External validation degradation. Performance gap between training and external test

Uncertainty calibration. Brier score, calibration curves

Latent space interpretability. Cluster separability by severity; qualitative review

Reproducibility. Can independent runner reproduce results from GitHub?

Publications. Preprints submitted, papers accepted, citation counts

Business Metrics

Partnership conversations. Number of active discussions per quarter

Letter of intent / pilots. Signed agreements (paid or unpaid)

Revenue. ARR or per-project (Year 2+)

Funding raised. Non-dilutive + dilutive cumulative

Advisor / co-founder pipeline. Active conversations with potential team members

Press / community recognition. Articles, conference talks, social engagement

Execution Metrics

GitHub commit cadence. Public progress signal; consistency matters

Experiments run per week. Indicator of research velocity

Documentation completeness. Can someone else understand and continue the work?

Sprint completion rate. Are we hitting our own milestones?

Hours invested per week. Sustainable pace tracking; burnout prevention

Learning velocity. New skills, techniques, tools mastered per month

Quarterly OKR Framework

Every quarter we set 3-5 OKRs (Objectives + Key Results). Each Objective is a qualitative goal. Each Key Result is a measurable outcome. We review weekly, score at quarter-end, and use scores to inform the next quarter's plan. The detailed execution documents will contain the actual quarterly OKRs.

Part VI — Beyond

18. Long-Term Vision (Years 4-7)

If Years 1-3 succeed, Years 4-7 are about scale, depth, and category leadership. We become the disease dynamics infrastructure that pharma, biotech, and research rely on. This vision is intentionally ambitious — we may not reach it. But it is the trajectory we are aiming at.

Year 4 — Platform Scale

- 5+ validated disease models (DR, AMD, glaucoma, diabetic nephropathy, plus one)
- \$10M+ ARR
- 20-40 person team
- Series B closed (\$25-50M)
- Multiple pharma multi-year contracts
- Open-source community around disease dynamics

Year 5 — Category Definition

- Recognized category leader in computational disease dynamics
- \$25-50M ARR
- International expansion (EU, Asia)
- First regulatory acceptance of synthetic cohorts in submissions
- Acquisition offers or IPO discussions
- Valuation \$500M – \$1B+ range

Years 6-7 — Outcome

- **Path A — Independent unicorn:** \$1B+ valuation, IPO trajectory, category leader
- **Path B — Strategic acquisition:** Acquired by pharma giant for \$1-5B, integrated into R&D;
- **Path C — Continued private:** Profitable, growing, optionality preserved
- **Path D — Partial exit:** Strategic minority investment from pharma, founders retain control

The Ultimate Mission

If Synapse fully succeeds, we contribute to a fundamental shift in how medicine thinks about disease: from static labels to dynamic processes, from observations to inferred state, from population statistics to personalized trajectories. We help drugs get approved faster, rare diseases get treated, and researchers understand disease evolution. That outcome is worth pursuing even if we only achieve part of it.

19. Glossary & References

Key Terms

Latent state-space model. A model that represents a system as having hidden internal states that evolve over time according to learned dynamics.

JEPA (Joint Embedding Predictive Architecture). A self-supervised learning approach (Yann LeCun et al.) that predicts representations of future inputs without reconstructing them.

HAWVA. Our framework name for the multimodal evidence fusion layer that maintains belief over hidden disease state.

Neural ODE / Latent ODE. Neural networks that learn continuous-time dynamics, useful for irregular time series like medical visits.

RETFound. Retinal foundation model trained on 1.6M unlabeled retinal images, available publicly via Hugging Face.

DR (Diabetic Retinopathy). Eye disease caused by diabetes; leading cause of blindness in working-age adults; our initial disease focus.

DME (Diabetic Macular Edema). Sub-condition of DR involving fluid leakage in the macula.

ETDRS scale. Early Treatment Diabetic Retinopathy Study scale; standard 0-4 severity grading for DR.

EyePACS. Public DR dataset on Kaggle (~35K labeled retinal images).

Messidor / Messidor-2. Smaller public retinal datasets for external validation.

DRCR Retina Network. US-based research consortium running protocolized DR/DME clinical trials.

Counterfactual. What would have happened under a different intervention/treatment.

Physician policy. The decision-making patterns of doctors that confound observational data.

Synthetic cohort. An artificially generated patient population that preserves real disease dynamics.

Active inference. Choosing what data to collect to maximally reduce uncertainty.

SaMD. Software as a Medical Device — FDA regulatory category for medical AI.

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- Bica, I., et al. (2020). Estimating Counterfactual Treatment Outcomes over Time Through Adversarially Balanced Representations.
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- Various ophthalmology references — see ongoing literature in living bibliography in repo.

Closing

This document is alive. It will be revised as evidence accumulates. Each tactical execution document — sprint plans, quarterly OKRs, technical specs — references back to this north star.

The discipline: *We hold two thoughts at once.*

First: We are attempting to build something genuinely valuable to medicine — infrastructure for understanding how disease evolves through time.

Second: We have not yet earned the right to claim we will succeed. Every claim must be evidence-backed. Every assertion must be honest.

The mission: *Build disease dynamics. Save lives. Earn the right to be a unicorn.*